

Speed of Thought

Also Remember...

- If you are on a very tight budget and only need basic office productivity, you should go for the low-cost Cyrix MIII processor from VIA. This works with any Socket 370 motherboard.
- If you play a lot of games you need to look for a processor that supports special instruction sets like 3DNow! (Athlon) or SSE (Pentium III/4).
- For high-end 3D design and CAD applications, you will require a processor with the highest clock speed you can afford and the largest amount of cache on board. This would make the Pentium and Athlon processors your best choice.

Cost and performance are the parameters that determine whether a processor fits into the ‘budget’ solution category or is an out and out ‘performance’ solution. AMD, the first to break the 1 GHz clock-speed barrier, has recently stolen a march over Intel as far as pure performance goes. This holds true for both the budget and performance categories, where the Duron and the Athlon outperform their equivalent and in some cases even higher-speed Intel counterparts.

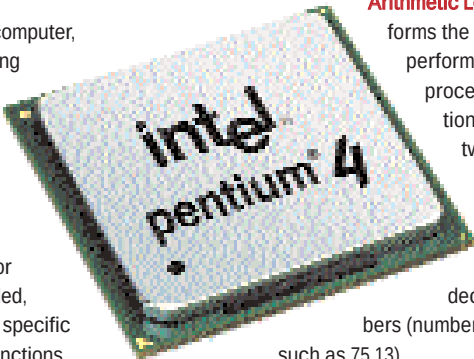
The prices of mainstream 32-bit processors have fallen substantially and you can now buy an Athlon 1.4 GHz processor for about Rs 9,500 giving you performance equivalent of a Pentium 4 1.8 GHz which retails at around Rs 12,500.

TECHNOLOGY INSIGHT

Everything that goes on inside a computer, from copying a simple file to playing an advanced computer game, is composed of a stream of binary data. The processor is a digital device that receives information from software, processes it in a predetermined fashion and returns the results. For this reason, the processor is divided, physically and electronically, into specific regions that perform dedicated functions.

A specific fabrication process is used to etch the die of a processor onto a silicon wafer. With the P-III processors, a 0.18 micron process is used. This will soon move to a 0.13 micron process which will allow 57 million transistors on a P4 processor.

Cache: There are two types of cache—data cache and instruction cache. After information is received by the processor, it is sent to a small temporary storage region (known as Level 1 cache) 64 bits at a time. Most modern processors have 256 K of Level 2 cache at the core frequency of the processor itself, thus providing a fast data path.



Arithmetic Logic Unit (ALU): This unit forms the core of the processor and performs the actual mathematical processing on the data like addition and subtraction. There are two ALUs, each of which handles two distinct types of numeric data types: integers (whole numbers that do not include a decimal) or floating-point numbers (numbers that involve decimals, such as 75.13).

Branch Prediction Unit: This is a very special part of the processor that inspects the nature of the instructions in the instruction cache and decides which of the ALUs is best suited for processing the data fastest. This branching of the data is done to ensure that no ALU is idle while the other is processing an instruction. This streamlines the decoding process and increases the efficiency of the processor.

Floating Point Unit (FPU): The Floating Point Unit is responsible for handling any mathematical operations involving decimal number.

	General desktop use	Home entertainment	Gaming and graphics professionals
You need	Decent performance in office productivity applications	Lag-free performance in all graphic, multimedia, video oriented applications	Movies, music, gaming and complete home entertainment with oodles of processing power
Look for	Any Celeron/Duron class processor with speeds in excess of 700 MHz	A Pentium III/Duron processor with speeds in excess of 900 MHz	An Athlon/Pentium 4 class processor with speeds in excess of 1.2 GHz
Our pick	Duron 800, Celeron 733 and above	Pentium III 1 GHz, Duron 900 and above	Athlon 1.4 GHz, Pentium 4 1.8 GHz
Price range	Up to Rs 5,000	Up to Rs 8,000	Rs 8,000 and above



The Top Performers on the Toughest Benchmarks

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